SIMATS Online Programming Lab

PYTHON PROGRAMMING

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Q1 **Harshad number**

10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: 18 → digit sum = 9, 18 % 9 == 0. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

```
n = int(input("Enter a number: "))

def is_harshad(num):
    s = sum(int(digit) for digit in str(num))
    return num % s == 0
```

Save Answer

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Q1 **Harshad number**

10 marks

A number is called a Harshad number (or Niven number) if it is divisible by the sum of its digits. Example: 18 → digit sum = 9, 18 % 9 == 0. Write a Python program to read a number N from the user and check whether it is a Harshad number. Also print all Harshad numbers between 1 and 500.

Your Answer

```
if n > 0 and is_harshad(n):
    print(f"{n} is a Harshad number.")
else:
    print(f"{n} is NOT a Harshad number.")

print("Harshad numbers between 1 and 500 are:")
```

Save Answer

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Q1 **Harshad number**

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Your Answer

```
    print(f"{n} is NOT a Harshad number.")

print("Harshad numbers between 1 and 500 are:")
for i in range(1, 501):
    if is_harshad(i):
        print(i, end=" ")
```

Save Answer

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Q2 GCD

10 marks

The following Python program is intended to find the GCD (Greatest Common Divisor) of two numbers using the Euclidean algorithm. It contains FOUR errors. Identify each error and write the corrected line.

a = int(input("Enter first: ")) b = int(input("Enter second: "))

while b = 0: temp = a a = b b = temp % a

print("GCD is", a)

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```python
a = int(input("Enter first: "))
b = int(input("Enter second: "))

while b != 0:
    temp = a
    a = b
    b = temp % a

print("GCD is", a)
```

Test Input

```
48
18
```

Output

```
Enter first: Enter second: GCD is 6
```

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Complete the missing lines in the Python program below. The program finds all pairs of elements in a list that sum to a target value. Fill in all seven blanks.

def find_pairs(lst, target):

seen = _____ pairs = []

for num in _____:

complement = target - _____

if complement in _____:

pairs.append((complement, num))

_____.add(num)

return _____

data = [int(x) for x in input("Enter numbers: ").split()]

t = int(input("Target: "))

result = _____(data, t)

print("Pairs:", result)

Your Answer

```
def find_pairs(lst, target):
    seen = set()
    pairs = []
    for num in lst:
        complement = target - num
        if complement in seen:
```

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Q4 **Bitwise Operators**

10 marks

Trace the following Python program step by step. Write the exact output. Also state the data type of each variable after assignment.

```
a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
print(f"a={a}, b={b}, c={c}, result={result}")
print(bin(result), oct(result), hex(result))
print(result >> 2, result << 1, ~result)
```

Your Answer

```
# Bitwise Operators - Trace and exact output

a = 0b1101
b = 0o25
c = 0x1A
result = (a & b) | c
```

Save Answer

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Your Fixed Code

```
def rotated_binary_search(arr, target):
    low = 0
    high = len(arr) - 1

    while low <= high:
        mid = (low + high) // 2

        if arr[mid] == target:
            return mid

        # Left half is sorted
        if arr[low] <= arr[mid]:
            if arr[low] <= target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        # Right half is sorted
        else:
            if arr[mid] < target <= arr[high]:
                low = mid + 1
            else:
                high = mid - 1

    return -1


data = [4, 5, 6, 7, 0, 1, 2]
```

Test Input

```
0
```

Output

```
Target: Found at: 4
```

Pending manual review

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```python

        if arr[mid] == target:
            return mid

        # Left half is sorted
        if arr[low] <= arr[mid]:
            if arr[low] <= target < arr[mid]:
                high = mid - 1
            else:
                low = mid + 1
        # Right half is sorted
        else:
            if arr[mid] < target <= arr[high]:
                low = mid + 1
            else:
                high = mid - 1

    return -1


data = [4, 5, 6, 7, 0, 1, 2]
t = int(input("Target: "))
result = rotated_binary_search(data, t)
print("Found at:", result)
```

```
Target: Found at: 4
```

Pending manual review

Submitted